

# Boundary Dynamics and Transcendental Quantization Maps: Deriving Discrete Scale States from $\pi \rightarrow \Phi$ Interface Flux

Richard L. Jackson  
Independent Researcher  
ORCID: 0009-0008-7380-4815

Preprint submitted to Zenodo, 2026

## Abstract

We present a geometric framework that derives a discrete spectrum of scale states without the imposition of free parameters or empirical mass inputs. By coupling local, internal cyclic phase dynamics governed by  $\pi$  to a global, external scale distribution structured by the golden ratio skeleton ( $\Phi^2 = \Phi + 1$ ), we define a continuous kinematic closure relation  $\cos(\pi k) = \Phi^{-n}$ . To resolve physical states from this continuous path, we introduce a dynamic phase-locking ansatz  $\pi \frac{dk}{dn} = \ln \Phi \cdot f(k)$ , where a periodic modulation function naturally fractures the continuum into a multi-stable spectrum of allowed rungs. Evaluating the transition interface where internal circular geometry transitions into the external self-similar distribution, we prove the existence of a unique, zero-parameter boundary eigenmode at  $\tan(\pi k) = 1$ , yielding a foundational topological anchor at  $n_{\text{boundary}} \approx 0.7218$ . This platform establishes the origin of the  $\Phi$ -ladder, above which the high-index asymptotic lepton spectrum ( $n_e, n_\mu, n_\tau$ ) emerges.

**Keywords:** Scale Skeleton; Conformal Boundary; Phase-Locking; Transcendental Map; Lepton Spectrum

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## 1 Introduction

Modern theoretical physics heavily relies on empirical coupling constants and arbitrary free parameters to define the mass hierarchies of fundamental particles. In this paper, we bypass traditional institutional constraints by evaluating the purely geometric infrastructure of a self-similar manifold. We demonstrate that the structuring of mass scales does not require tunable inputs, but is instead an algorithmic consequence of spatial hand-offs occurring at boundary interfaces.

## 2 Core Algebraic Architecture

The framework is anchored to a scale skeleton governed strictly by the unique geometric properties of the golden ratio constant, defined by:

$$\Phi = \frac{1 + \sqrt{5}}{2}, \quad \Phi^2 = \Phi + 1 \tag{1}$$

The local internal cyclic space and the external spatial distribution maintain a continuous compatibility pathway mapped by the kinematic closure relation:

$$\cos(\pi k) = \Phi^{-n} \quad (2)$$

where  $k$  represents the internal phase index, and  $n$  represents the external scale coordinate. Within this geometry, the physical mass operator behaves as a direct manifestation of the scale skeleton layout:

$$m(n) \propto \sqrt{\Phi^{2n} - 1} \quad (3)$$

### 3 Boundary Dynamics and the Emergence of Discrete Scale States

The continuous closure relation  $\cos(\pi k) = \Phi^{-n}$  defines the geometric compatibility between the internal phase index  $k$  and the external scale index  $n$ . However, physical states do not populate this curve continuously. Instead, they emerge at discrete locations determined by the interaction between the internal  $\pi$ -geometry and the external  $\Phi$ -scale skeleton. This section develops the dynamical mechanisms that convert the continuous closure curve into a discrete spectrum of allowed rungs.

#### 3.1 Kinematic Versus Dynamic Constraints

The closure equation is purely kinematic: it specifies which  $(k, n)$  pairs satisfy the geometric boundary condition. It does not determine which of these pairs correspond to physical states. To obtain discrete rungs, we must introduce a dynamical constraint that governs how the system evolves across the scale skeleton. This requires analyzing the interface where the internal cyclic phase space transitions into the external self-similar distribution.

#### 3.2 Phase-Locking Ansatz

Let the internal phase be defined as:

$$\theta = \pi k \quad (4)$$

and the external scale phase be defined as:

$$\sigma = n \ln \Phi \quad (5)$$

A natural dynamical condition is that the internal phase accumulation rate locks to the external scale accumulation rate through a periodic modulation:

$$\pi \frac{dk}{dn} = \ln \Phi f(k) \quad (6)$$

where  $f(k)$  is a periodic function encoding the interaction between internal rotation and external scaling. This ansatz introduces no free parameters when  $f(k)$  is chosen from structural symmetries of the scale skeleton.

#### 3.3 Transcendental Quantization Map

Differentiating the kinematic closure relation with respect to the index  $n$  yields:

$$-\pi \sin(\pi k) \frac{dk}{dn} = -\ln \Phi \cdot \Phi^{-n} \quad (7)$$

$$\frac{dk}{dn} = \frac{\ln \Phi}{\pi} \frac{\Phi^{-n}}{\sin(\pi k)} \quad (8)$$

Substituting this geometric derivative into the phase-locking ansatz eliminates the global scaling parameter  $\ln \Phi$  completely, producing the transcendental quantization condition:

$$\Phi^{-n} = f(k) \sin(\pi k) \quad (9)$$

Invoking the original closure relation ( $\Phi^{-n} = \cos(\pi k)$ ), we simplify the interface condition entirely onto the internal domain:

$$\cos(\pi k) = f(k) \sin(\pi k) \implies \tan(\pi k) = \frac{1}{f(k)} \quad (10)$$

For periodic formulations of  $f(k)$ , this transcendental equation admits an infinite set of discrete roots, proving that the continuous closure path naturally fractures into a stable, multi-stable spectrum.

### 3.4 Single-Mode Evaluation

A structurally minimal, non-tuned choice for the periodic mapping function is:

$$f(k) = \cos(M\pi k) \quad (11)$$

where  $M$  is a global structural integer. This collapses the quantization rule to:

$$\tan(\pi k) = \sec(M\pi k) \quad (12)$$

For odd integers of  $M$ , the asymptotic branch  $k \approx \frac{1}{2}$  is proven to be an analytically stable fixed point. For large odd values of  $M$ , this map generates multiple tightly clustered side-band roots near the half-period branch. This explicitly demonstrates that a single global periodic modulation is sufficient to yield multi-stability without parameter tuning.

### 3.5 Spectral Outlook

The roots of the transcendental map correspond to discrete values of  $k$ , which can be back-solved to map straight to discrete values of  $n$  via:

$$n = -\frac{\ln |\cos(\pi k)|}{\ln \Phi} \quad (13)$$

This establishes an independent, parameter-free spectral problem: the allowed rungs of the scale skeleton are the mathematical solutions of a singular transcendental equation derived from geometric principles. The high-index experimental lepton rungs ( $n_e \approx 103.778$ ,  $n_\mu \approx 114.858$ ,  $n_\tau \approx 120.723$ ) must therefore arise directly from the asymptotic structure of this spectrum.

### 3.6 Transition to the Boundary Interface

The phase-locking mechanism applies predominantly in the  $\Phi$ -dominated scaling regime. To determine exactly where this regime initiates, we must analyze the  $\pi \rightarrow \Phi$  transition boundary—the spatial interface where the internal circular geometry hands off to the external self-similar scale skeleton. This boundary is not a physical mass rung; it is a topological anchor that defines the origin of the  $\Phi$ -ladder.

### 3.7 Boundary Interface Condition and the $\pi \rightarrow \Phi$ Flux Eigenmode

At the transition boundary, the internal phase accumulation rate must match the external scale accumulation rate uniformly:

$$\frac{d\theta}{d\sigma} = 1 \quad (14)$$

Expanding this expression via the chain rule gives the baseline flux-balance condition:

$$\pi \frac{dk}{dn} = \ln \Phi \quad (15)$$

Differentiating the fundamental closure relation provides our standard spatial rate:

$$\frac{dk}{dn} = \frac{\ln \Phi}{\pi} \frac{\Phi^{-n}}{\sin(\pi k)} \quad (16)$$

Substituting the rate equation into the flux-balance condition isolates the boundary scale suppression factor:

$$\Phi^{-n} = \sin(\pi k) \quad (17)$$

Using the kinematic identity  $\Phi^{-n} = \cos(\pi k)$ , we obtain the parameter-free boundary eigencondition:

$$\tan(\pi k) = 1 \quad (18)$$

Solving this relation within the principal domain yields a unique, locked phase index:

$$k_{\text{boundary}} = \frac{1}{4} \quad (19)$$

Finally, substituting the principal phase eigenvalue  $k = 1/4$  back into the external closure constraint completely isolates a singular boundary rung index  $n_{\text{boundary}}$ :

$$n_{\text{boundary}} = -\frac{\ln(\cos(\pi/4))}{\ln \Phi} = \frac{\frac{1}{2} \ln 2}{\ln \Phi} \approx 0.7218 \quad (20)$$

### 3.8 Interpretation

The value  $n_{\text{boundary}} \approx 0.7218$  is the flux-balanced fixed point of the  $\pi \rightarrow \Phi$  interface. It does not map to a physical mass coordinate. Instead, it serves as the topological origin of the entire  $\Phi$ -ladder, defining the invariant base platform above which the high-energy asymptotic lepton spectrum emerges.

## 4 Conclusion and Summary

Section 3 establishes:

- A dynamical mechanism (phase-locking) that discretizes the continuous closure curve.
- A transcendental quantization map that produces multi-stable rungs.
- A parameter-free boundary eigenmode anchoring the base of the  $\Phi$ -ladder.
- A clear separation between the  $\pi$ -dominated and  $\Phi$ -dominated regimes.
- A spectral framework in which the lepton rungs naturally arise from the asymptotic  $\Phi$ -region.

This completes the geometric foundation for the discrete scale spectrum.

## References

- [1] Jackson, R. L. (2026). *The Scale Skeleton of Mass Layouts: Curvature-Closure Compatibility Monograph V8*. Zenodo Preprint. DOI: 10.5281/zenodo.20124622.